The Induction Principle — problem 8.31

A. Engel — Problem-Solving Strategies · 1998

(a) Consider any convex region in the plane crossed by l lines with p interior points of intersection. Find a simple relationship between l, p, and the number r of disjoint regions created.

(b) Place n distinct points on the circumference of a circle, and draw all possible chords through pairs of these points. Assume that no three chords are concurrent. Let a_n be the number of regions. Find a_1, a_2, a_3, a_4, a_5 by drawing figures. Guess a_n, and check your guess by finding a_6. Now find a_n by using the result in (a).